

Generalized Linear Models

In previous chapters, we have seen how to model a binomial or Poisson response. Multinomial response models can often be recast as Poisson responses and the standard linear model with a normal (Gaussian) response is already familiar. Although these models each have their distinctive characteristics, we see some common features in all of them. We can abstract these to form the *generalized linear model* (GLM). By developing a theory and constructing general methods for GLMs, we are able to tackle a wider range of data with different types of response variables. GLMs were introduced by Nelder and Wedderburn (1972), while McCullagh and Nelder (1989) provide a book-length treatment.

8.1 GLM Definition

A GLM is defined by specifying two components. The response should be a member of the exponential family distribution and the link function describes how the mean of the response and a linear combination of the predictors are related.

Exponential Family: In a GLM the distribution of Y is from the exponential family of distributions which take the general form:

$$f(y|\theta, \phi) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]$$

The θ is called the *canonical parameter* and represents the location, while ϕ is called the *dispersion parameter* and represents the scale. We may define various members of the family by specifying the functions a , b and c . The most commonly used examples are:

1. Normal or Gaussian:

$$\begin{aligned} f(y|\theta, \phi) &= \frac{1}{\sqrt{2\pi\sigma}} \exp \left[-\frac{(y-\mu)^2}{2\sigma^2} \right] \\ &= \exp \left[\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2} \left(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2) \right) \right] \end{aligned}$$

So we can write $\theta = \mu$, $\phi = \sigma^2$, $a(\phi) = \phi$, $b(\theta) = \theta^2/2$ and $c(y, \phi) = -(y^2/\phi + \log(2\pi\phi))/2$.

2. Poisson:

$$f(y|\theta, \phi) = e^{-\mu} \mu^y / y!$$

$$= \exp(y \log \mu - \mu - \log y!)$$

So we can write $\theta = \log(\mu)$, $\phi \equiv 1$, $a(\phi) = 1$, $b(\theta) = \exp(\theta)$ and $c(y, \phi) = -\log y!$.

3. Binomial:

$$\begin{aligned} f(y|\theta, \phi) &= \binom{n}{y} \mu^y (1-\mu)^{n-y} \\ &= \exp \left(y \log \mu + (n-y) \log(1-\mu) + \log \binom{n}{y} \right) \\ &= \exp \left(y \log \frac{\mu}{1-\mu} + n \log(1-\mu) + \log \binom{n}{y} \right) \end{aligned}$$

So we see that $\theta = \log \frac{\mu}{1-\mu}$, $b(\theta) = -n \log(1-\mu) = n \log(1 + \exp \theta)$ and $c(y, \phi) = \log \binom{n}{y}$.

The gamma and inverse Gaussian are other less used members of the exponential family that are covered in Chapter 9. Notice that in the normal density, the ϕ parameter is free (as it is also for the gamma density), while for the Poisson and binomial it is fixed at one. This is because the Poisson and binomial are one-parameter families, while the normal and gamma have two parameters. In fact, some authors reserve the term *exponential family* distribution for cases where ϕ is not used, while using the term *exponential dispersion family* for cases where it is. This has important consequences for the analysis.

Some other densities, such as the negative binomial and the Weibull distribution, are not members of the exponential family, but they are sufficiently close that the GLM can be fit with some modifications. It is also possible to fit distributions that are not in the exponential family using the GLM-style approach, but there are some additional complications.

We now derive the mean and variance of the exponential family distributions. The log-likelihood for a single y is given by:

$$l(\theta) = (y\theta - b(\theta))/a(\phi) + c(y, \phi)$$

Taking derivatives with respect to θ gives:

$$l'(\theta) = (y - b'(\theta))/a(\phi)$$

Taking the expectation over y gives:

$$El'(\theta) = (EY - b'(\theta))/a(\phi)$$

From general likelihood theory, we know that $El'(\theta) = 0$ at the true value of θ which implies that:

$$EY = \mu = b'(\theta)$$

Now taking second derivatives:

$$l''(\theta) = -b''(\theta)/a(\phi)$$

General likelihood theory tells us that $El''(\theta) = -E[(l'(\theta))^2]$. We evaluate at the true value of θ to obtain

$$b''(\theta)/a(\phi) = E[(Y - b'(\theta))^2]/a^2(\phi)$$

which gives us

$$\text{var } Y = b''(\theta)a(\phi)$$

The mean is a function of θ only, while the variance is a product of functions of the location and the scale. $V(\mu) = b''(\theta)/w$ is called the *variance function* and describes how the variance relates to the mean using the known relationship between θ and μ .

In the Gaussian case, $b''(\theta) = 1$ and so the variance is independent of the mean. For other distributions, this is not true, making the Gaussian case exceptional. We can introduce weights by setting:

$$a(\phi) = \phi/w$$

where w is a known weight that varies between observations. We could allow for general choices of a but the simplified form, including weights if needed, covers all the situations we currently need.

Link Function: Suppose we may express the effect of the predictors on the response through a *linear predictor*:

$$\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = x^T \beta$$

The link function, g , describes how the mean response, $EY = \mu$, is linked to the covariates through the linear predictor:

$$\eta = g(\mu)$$

In principle, any monotone continuous and differentiable function will do, but there are some convenient and common choices for the standard GLMs.

In the Gaussian linear model, the identity link, $\eta = \mu$, is the obvious selection, but another choice would give $y = g^{-1}(x^T \beta) + \varepsilon$. This does not correspond directly to a transform on the response: $g(y) = x^T \beta + \varepsilon$ as, for example, in a Box–Cox type transformation. In a GLM, the link function is assumed known whereas in a *single index model*, g is estimated.

For the Poisson GLM, the mean μ must be positive so $\eta = \mu$ will not work conveniently since η can be negative. The standard choice is $\mu = e^\eta$, so that $\eta = \log \mu$ which ensures $\mu > 0$. This log link means that additive effects of x lead to multiplicative effects on μ .

For the binomial GLM, let p be the probability of success and let this be our μ if we define the response as the proportion rather than the count. This requires that $0 \leq p \leq 1$. There are several commonly used ways to ensure this: the logistic, probit and complementary log-log links. These are discussed in detail in Chapter 2.

The *canonical link* has g such that $\eta = g(\mu) = \theta$, the canonical parameter of the exponential family distribution. This means that $g(b'(\theta)) = \theta$. The canonical links

Family	Link	Variance Function
Normal	$\eta = \mu$	1
Poisson	$\eta = \log \mu$	μ
Binomial	$\eta = \log(\mu/(1-\mu))$	$\mu(1-\mu)$
Gamma	$\eta = \mu^{-1}$	μ^2
Inverse Gaussian	$\eta = \mu^{-2}$	μ^3

Table 8.1 *Canonical links for GLMs.*

for the common GLMs are shown in Table 8.1. If a canonical link is used, $X^T Y$ is *sufficient* for β . The canonical link is mathematically and computationally convenient and is often the natural choice of link. However, one is not required to use the canonical link and sometimes context may compel another choice.

8.2 Fitting a GLM

The parameters, β , of a GLM can be estimated using maximum likelihood. The log-likelihood for a single observation, where $a_i(\phi) = \phi/w_i$, is:

$$l(\beta; y_i) = w_i \left[\frac{y_i \theta_i - b(\theta_i)}{\phi} \right] + c(y_i, \phi)$$

So for independent observations, the log-likelihood will be $\sum_i l(\beta; y_i)$. We want to maximize this over β and start by taking partial derivatives with respect to the components β_j :

$$\frac{\partial l}{\partial \beta_j} = \frac{1}{\phi} \sum_i w_i \left(y_i \frac{\partial \theta_i}{\partial \beta_j} - b'(\theta_i) \frac{\partial \theta_i}{\partial \beta_j} \right)$$

The chain rule gives us:

$$\frac{\partial \theta_i}{\partial \beta_j} = \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \beta_j}$$

Using the fact that $\frac{\partial \mu_i}{\partial \theta_i} = b''(\theta_i)$, we have

$$\frac{\partial l}{\partial \beta_j} = \frac{1}{\phi} \sum_i \frac{(y_i - b'(\theta_i))}{b''(\theta_i)/w_i} \frac{\partial \mu_i}{\partial \beta_j}$$

We now substitute in the known relations for the mean and variance functions. We set the partial derivatives to zero to obtain the maximum likelihood estimates as the solution to:

$$\sum_i \frac{(y_i - \mu_i)}{V(\mu_i)} \frac{\partial \mu_i}{\partial \beta_j} = 0 \quad \forall j$$

Now suppose we knew the variance function $V(\mu)$, then we would obtain this same set of equations if we had started out to minimize the weighted least squares criterion:

$$\sum_i \frac{(y_i - \mu_i)^2}{V(\mu_i)}$$

This suggests how we might develop an algorithm for deriving the estimates using iteratively reweighted least squares (IRWLS). Full details can be found in McCullagh and Nelder (1989) or Wood (2006).

The procedure requires an initial guess of $\hat{\mu}^0$ from which $\hat{\eta}^0$ where the superscript indicates the iteration number.

1. Form the “adjusted dependent variable” $z^i = \hat{\eta}^i + (y - \hat{\mu}^i) \frac{d\eta}{d\mu} \big|_{\hat{\eta}^i}$.
2. Form the weights $1/w^0 = \left(\frac{d\eta}{d\mu} \right)^2 \big|_{\hat{\eta}^0} V(\hat{\mu}^0)$.
3. Reestimate to get $\hat{\beta}^{i+1}$ and hence $\hat{\eta}^{i+1}$.

Repeat these steps until convergence.

Notice that the fitting procedure uses only $\eta = g(\mu)$ and $V(\mu)$, but requires no further knowledge of the distribution of y . This point will be important later when considering the quasi-likelihood method in Section 9.4. Estimates of variance may be obtained using standard likelihood theory from:

$$\text{var}(\hat{\beta}) = (X^T W X)^{-1} \hat{\phi}$$

where W is a diagonal matrix formed from the weights w . This is comparable to the form used in weighted least squares with the exception that the weights are now a function of the response for a GLM.

Let’s implement the procedure explicitly to understand how the fitting algorithm works. We use the Bliss data from Section 4.2 to illustrate this. Here is the fit we are trying to match:

```
data(bliss, package="faraway")
mod1 <- glm(cbind(dead,alive) ~ conc, family=binomial, bliss)
summary(mod1)$coef
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.3238	0.41789	-5.5608	2.6854e-08
conc	1.1619	0.18142	6.4046	1.5077e-10

For a binomial response, we have:

$$\eta = \log \frac{\mu}{1-\mu} \quad \frac{d\eta}{d\mu} = \frac{1}{\mu(1-\mu)} \quad V(\mu) = \mu(1-\mu)/n \quad w = n\mu(1-\mu)$$

where the variance is computed with the understanding that y is the proportion, not the count. We use y for our initial guess for $\hat{\mu}$ which works here because none of the observed proportions are zero or one:

```
y <- bliss$dead/30; mu <- y
library(faraway)
eta <- logit(mu)
z <- eta + (y-mu)/(mu*(1-mu))
w <- 30*mu*(1-mu)
lmod <- lm(z ~ conc, weights=w, bliss)
coef(lmod)
```

	conc
(Intercept)	-2.3025
	1.1536

It is interesting how close these initial estimates are to the converged values given above. This is not uncommon. Even so, to get a more precise result, iteration is necessary. We do five iterations here:

```

for(i in 1:5){
  eta <- lmod$fit
  mu <- ilogit(eta)
  z <- eta + (y-mu)/(mu*(1-mu))
  w <- 30*mu*(1-mu)
  lmod <- lm(z ~ bliss$conc, weights=w)
  cat(i,coef(lmod),"\\n")
}
1 -2.3237 1.1618
2 -2.3238 1.1619
3 -2.3238 1.1619
4 -2.3238 1.1619
5 -2.3238 1.1619

```

We can see that convergence is fast in this case. In most cases, the convergence is rapid. If there is a failure to converge, this is often a sign of some problem with the model specification or an unusual feature of the data. An example of such a problem with the estimation may be seen in Section 2.7. A look at the final (weighted) linear model reveals that:

```
summary(lmod)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.3238	0.1462	-15.9	0.00054
bliss\$conc	1.1619	0.0635	18.3	0.00036

n = 5, p = 2, Residual SE = 0.350, R-Squared = 0.99

The standard errors are not correct and can be computed (rather inefficiently) as follows:

```

xm <- model.matrix(lmod)
wm <- diag(w)
sqrt(diag(solve(t(xm) %*% wm %*% xm)))
[1] 0.41787 0.18141

```

Now $\hat{\text{var}}(\hat{\beta}) = (X^T W X)^{-1}$ because $\phi = 1$ for the binomial model but in the Gaussian linear model $\hat{\text{var}}(\hat{\beta}) = (X^T W X)^{-1} \hat{\sigma}^2$. To get the correct standard errors from the `lm` fit, we need to scale out the $\hat{\sigma}$ as follows:

```
summary(lmod)$coef[,2]/summary(lmod)$sigma
```

	conc
(Intercept)	0.41789
	0.18142

These calculations are shown for illustration purposes only and are done more efficiently and reliably by the `glm` function.

8.3 Hypothesis Tests

When considering the choice of model for some data, we must define the range of possibilities. The *null* model is the smallest model we will entertain while the *full* or *saturated* model is the most complex.

The null model represents the situation where there is no relation between the predictors and the response. Usually this means we fit a common mean μ for all y , that is, one parameter only. For the Gaussian GLM, this is the model $y = \mu + \epsilon$. For some contingency table models, there will be additional parameters that represent row or column totals or other such constraints. In these cases, the null model will have more than one parameter.

In the saturated model, the response is explained in the best achievable way. There are usually several ways in which this can be done by adding sufficiently many parameters. One way that will always be available is to define a factor with one level for each unique combination of the predictors appearing in the data. In many datasets, there will be only one observation for each unique combination. In such cases there will be n parameters for n data points and $\hat{\mu} = y$. In other examples, where replication occurs, we will need fewer than n parameters.

A statistical model describes how we partition the data into systematic structure and random variation. The null model represents one extreme where the data is represented entirely as random variation, while the saturated or full model represents the data as being entirely systematic, excepting any variation that no model could explain.

We would like a measure of how well our model fits. We can do this by comparing our model to the full model. Hence we consider the difference between the log-likelihood for the full model, $l(y, \phi|y)$, and that for the model under consideration, $l(\hat{\mu}, \phi|y)$, expressed as a log likelihood ratio statistic:

$$2(l(y, \phi|y) - l(\hat{\mu}, \phi|y))$$

The factor of two is used for convenience in the subsequent distributional results. Provided that the observations are independent and for an exponential family distribution, when $a_i(\phi) = \phi/w_i$, this simplifies to:

$$\sum_i 2w_i(y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i))/\phi$$

where $\tilde{\theta}$ are the estimates under the full (saturated) model and $\hat{\theta}$ are the estimates under the model of interest. We write this quantity as $D(y, \hat{\mu})/\phi$ where $D(y, \hat{\mu})$ is called the *deviance* and $D(y, \hat{\mu})/\phi$ is called the *scaled deviance*. Deviances for the common GLMs are shown in Table 8.2.

GLM	Deviance
Gaussian	$\sum_i (y_i - \hat{\mu}_i)^2$
Poisson	$2 \sum_i [y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)]$
Binomial	$2 \sum_i [y_i \log(y_i/\hat{\mu}_i) + (m - y_i) \log((m - y_i)/(m - \hat{\mu}_i))]$
Gamma	$2 \sum_i [-\log(y_i/\hat{\mu}_i) + (y_i - \hat{\mu}_i)/\hat{\mu}_i]$
Inverse Gaussian	$\sum_i (y_i - \hat{\mu}_i)^2 / (\hat{\mu}_i^2 y_i)$

Table 8.2 For the binomial $y_i \sim B(m, p_i)$ and $\mu_i = mp_i$, that is, μ is the count and not proportion in this formula. For the Poisson, the deviance is known as the *G-statistic*. The second term $\sum_i (y_i - \hat{\mu}_i)$ is usually zero if an intercept term is used in the model.

An alternative measure of fit that is sometimes used in place of the deviance is the Pearson's X^2 statistic:

$$X^2 = \sum_i \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$$

where $V(\hat{\mu}) = \text{var}(\hat{\mu})$.

There are two main types of hypothesis test. The goodness of fit test asks whether the current model fits the data. The other type of test compares two nested models where the smaller model represents a linear restriction on the parameters of the larger model. The goodness of fit test can be viewed as a model comparison test if we identify the smaller model with the model of interest and the larger model with the full or saturated model.

For the goodness of fit test, we use the fact that, under certain conditions, provided the model is correct, the scaled Deviance and the Pearson's X^2 statistic are both asymptotically χ^2 with degrees of freedom equal to the number of observations minus the number of identifiable parameters. For GLMs with a dispersion parameter, such as the Gaussian, we usually do not know the value of the ϕ , and so this test usually cannot be used. For the binomial and the Poisson, $\phi = 1$, and so the test is practical. However, the accuracy of the asymptotic approximation is dubious for smaller datasets. For a binary response, the approximation is worthless as explained in Section 2.3.

For comparing a larger model, Ω , to a smaller nested model, ω , the difference in the scaled deviances, $D_\omega - D_\Omega$, is asymptotically χ^2 with degrees of freedom equal to the difference in the number of identifiable parameters in the two models. For the Gaussian model and other models where the dispersion ϕ is usually not known, this test cannot be directly used. However, if we insert an estimate of ϕ we may compute an F -statistic of the form:

$$\frac{(D_\omega - D_\Omega)/(df_\omega - df_\Omega)}{\hat{\phi}}$$

where $\hat{\phi} = X^2/(n - p)$ is a good estimate of the dispersion. The degrees of freedom df are typically the number of observations minus the number of parameters. For the Gaussian model, a sensible estimate is $\hat{\phi} = RSS_\Omega/df_\Omega$, and the resulting F -statistic has an exact F distribution for the null. For other GLMs with free dispersion parameters, the statistic is only approximately F distributed.

For every GLM except the Gaussian, an approximate null distribution must be used whose accuracy may be in doubt particularly for smaller samples. However, the approximation is better when comparing models than for the goodness of fit statistic.

Let's consider the possible tests on the Bliss insect data:

```
summary(mod1)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.324	0.418	-5.56	2.7e-08
conc	1.162	0.181	6.40	1.5e-10

```
n = 5 p = 2
```

```
Deviance = 0.379 Null Deviance = 64.763 (Difference = 64.385)
```

We are able to make a goodness of fit test by examining the size of the residual deviance compared to its degrees of freedom:

```
1-pchisq(deviance(mod1), df.residual(mod1))
```

```
[1] 0.9446
```

where we see the p -value is large indicating no evidence of a lack of fit. As with lack of fit tests for Gaussian linear models, this outcome does not mean that this model is correct or that no better models exist. We can also see that the null model would be

inadequate for the data since the null deviance of 64.7 is very large for four degrees of freedom.

We can test for the significance of the linear concentration term by comparing the current model to the null model:

```
anova(mod1, test="Chi")
```

Analysis of Deviance Table

Model: binomial, link: logit

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	P(> Chi)
NULL			4	64.8	
conc	1	64.4	3	0.4	1e-15

We see that the concentration term is clearly significant. We can also fit and test a more complex model:

```
modl2 <- glm(cbind(dead, alive) ~ conc + I(conc^2), family=binomial, bliss)
```

```
anova(mod1, modl2, test="Chi")
```

	Resid. Df	Resid. Dev	Df	Deviance	P(> Chi)
1	3	0.379			
2	2	0.195	1	0.183	0.669

We can see that there is no need for a quadratic term in the model. The same information could be extracted with:

```
anova(modl2, test="Chi")
```

We may also take a Wald test approach by taking the standard error of the parameter estimates to construct a z -statistic of the form $\hat{\beta}/se(\hat{\beta})$. This has an asymptotically normal null distribution. For the Bliss data, for the concentration term, we have $z = 1.162/0.181 = 6.40$. Thus the (approximate) p -value for the Wald test of the concentration parameter being equal to zero is $1.5e^{-10}$ and thus we clearly reject the null here. Remember that this is again only an approximate test except in the special case of the Gaussian GLM where the z -statistic is the t -statistic and has an exact t -distribution. The difference of deviances test is preferred to the Wald test, due, in part, to the problem noted by Hauck and Donner (1977).

8.4 GLM Diagnostics

As with standard linear models, it is important to check the adequacy of the assumptions that support the GLM. The diagnostic methods for GLMs mirror those used for Gaussian linear models. However, some adaptations are necessary and, depending on the type of GLM, not all diagnostic methods will be applicable.

Residuals: Residuals represent the difference between the data and the model and are essential to explore the adequacy of the model. In the Gaussian case, the residuals are $\hat{\epsilon} = y - \hat{\mu}$. These are called response residuals for GLMs, but since the variance of the response is not constant for most GLMs, some modification is necessary. We would like residuals for GLMs to be defined such that they can be used in a similar way as in the Gaussian linear model.

The *Pearson residual* is comparable to the standardized residuals used for linear models and is defined as:

$$r_P = \frac{y - \hat{\mu}}{\sqrt{V(\hat{\mu})}}$$

where $V(\mu) \equiv b''(\theta)$. These are just a rescaling of $y - \hat{\mu}$. Notice that $\sum r_p^2 = X^2$ and hence the name. Pearson residuals can be skewed for nonnormal responses.

The *deviance residuals* are defined by analogy to Pearson residuals. The Pearson residual was r_p such that $\sum r_p^2 = X^2$, so we set the deviance residual as r_D such that $\sum r_D^2 = \text{Deviance} = \sum d_i$. Thus:

$$r_D = \text{sign}(y - \hat{\mu})\sqrt{d_i}$$

For example, in the Poisson:

$$r_D = \text{sign}(y - \hat{\mu})[2(y \log y / \hat{\mu} - y + \hat{\mu})]^{1/2}$$

Let's examine the types of residuals available to us using the Bliss data. We can obtain the deviance residuals as:

```
residuals(modl)
[1] -0.451015  0.359696  0.000000  0.064302 -0.204493
```

These are the default choice of residuals. The Pearson residuals are:

```
residuals(modl, "pearson")
      1      2      3      4      5
-0.432523  0.364373  0.000000  0.064147 -0.208107
```

which are just slightly different from the deviance residuals. The response residuals are:

```
residuals(modl, "response")
      1      2      3      4      5
-0.0225051  0.0283435  0.0000000  0.0049898 -0.0108282
```

which is just the response minus the fitted value:

```
bliss$dead/30 - fitted(modl)
      1      2      3      4      5
-0.0225051  0.0283435  0.0000000  0.0049898 -0.0108282
```

Finally, the so-called working residuals are:

```
residuals(modl, "working")
      1      2      3      4      5
-0.277088  0.156141  0.000000  0.027488 -0.133320
```

```
modl$residuals
      1      2      3      4      5
-0.277088  0.156141  0.000000  0.027488 -0.133320
```

Note that it is important to use the `residuals()` function to get the deviance residuals which are most likely what is needed for diagnostic purposes. Using `$residuals` gives the working residuals which is not usually needed for diagnostics. We can now identify the working residuals as a by-product of the IRWLS fitting procedure from Section 8.2.

```
residuals(lmod)
      1      2      3      4      5
-2.7709e-01  1.5614e-01 -3.8463e-16  2.7488e-02 -1.3332e-01
```

Leverage and Influence: For a linear model, $\hat{y} = Hy$, where $H = X(X^T X)^{-1} X^T$ is the *hat matrix* that projects the data onto the fitted values. The leverages h_i are given by the diagonal of H and represent the potential of the point to influence the fit. They are solely a function of X and whether they are in fact influential will also depend on y . Leverages are somewhat different for GLMs. The IRWLS algorithm used to fit the GLM uses weights, w . These weights are just part of the IRWLS

algorithm and are not user assigned. However, these do affect the leverage. We form a matrix $W = \text{diag}(w)$ and the hat matrix is:

$$H = W^{1/2}X(X^T W X)^{-1}X^T W^{1/2}$$

We extract the diagonal elements of H to get the leverages h_i . A large value of h_i indicates that the fit may be sensitive to the response at case i . Large leverages typically mean that the predictor values are unusual in some way. One important difference from the linear model case is that the leverages are no longer just a function of X and now depend on the response through the weights W . The leverages may be calculated as:

```
influence(mod1) $hat
```

```
      1      2      3      4      5
0.42550 0.41331 0.32238 0.41331 0.42550
```

As in the linear model case, we might choose to studentize the residuals as follows:

$$r_{SD} = \frac{r_D}{\sqrt{\hat{\phi}(1 - h_i)}}$$

or compute jackknife residuals representing the difference between the observed response for case i and that predicted from the data with case i excluded, scaled appropriately. These are expensive to compute exactly and so an approximation due to Williams (1987) can be used:

$$\text{sign}(y - \hat{\mu}) \sqrt{(1 - h_i) r_{SD}^2 + h_i r_{SP}^2}$$

where $r_{SP} = r_P / \sqrt{1 - h_i}$. These may be computed as:

```
rstudent(mod1)
```

```
      1      2      3      4      5
-0.584786 0.472135 0.000000 0.083866 -0.271835
```

Outliers may be detected by observing particularly large jackknife residuals.

Leverage only measures the potential to affect the fit whereas measures of influence more directly assess the effect of each case on the fit. We can examine the change in the fit from omitting a case by looking at the changes in the coefficients:

```
influence(mod1) $coef
```

```
      (Intercept)      conc
1  -0.2140015    0.0806635
2   0.1556719   -0.0470873
3   0.0000000    0.0000000
4  -0.0058417    0.0084177
5   0.0492639   -0.0365734
```

Alternatively, we can examine the Cook statistics:

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^T (X^T W X) (\hat{\beta}_{(i)} - \hat{\beta})}{p \hat{\phi}}$$

which may be calculated as:

```
cooks.distance(mod1)
```

1 2 3 4 5
 0.1205927 0.0797100 0.0000000 0.0024704 0.0279174

We can see that the biggest change would occur by omitting the first observation. However, since this is a very small dataset with just five observations, we would not contemplate dropping cases. In any event, we see that the change in the coefficients would not qualitatively change the conclusion.

Model Diagnostics: We may divide diagnostic methods into two types. Some methods are designed to detect single cases or small groups of cases that do not fit the pattern of the rest of the data. Outlier detection is an example of this. Other methods are designed to check the assumptions of the model. These methods can be subdivided into those that check the structural form of the model, such as the choice and transformation of the predictors, and those that check the stochastic part of the model, such as the nature of the variance about the mean response. Here, we focus on methods for checking the assumptions of the model.

For linear models, the plot of residuals against fitted values is probably the single most valuable graphic. For GLMs, we must decide on the appropriate scale for the fitted values. Usually, it is better to plot the linear predictors $\hat{\eta}$ rather than the predicted responses $\hat{\mu}$. We revisit the model for Galápagos data first presented in Section 5.1. Consider first a plot using $\hat{\mu}$ presented in the first panel of Figure 8.1:

```
data(gala, package="faraway")
gala <- gala[,-2]
modp <- glm(Species ~ ., family=poisson, gala)
plot(residuals(modp) ~ predict(modp, type="response"),
     xlab=expression(hat(mu)), ylab="Deviance residuals")
```

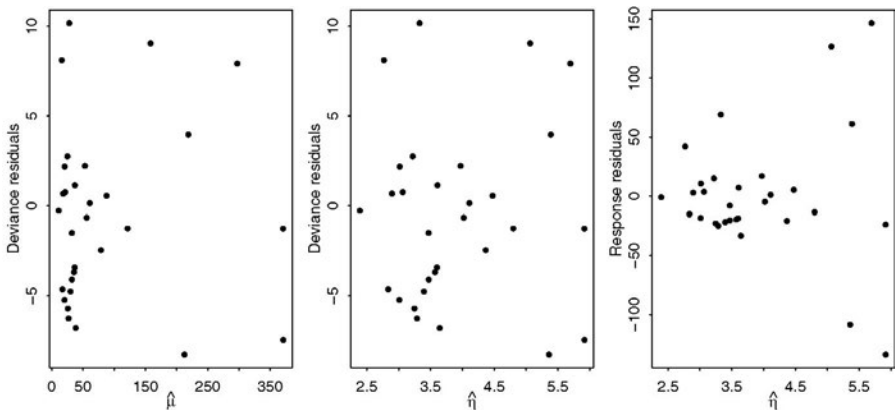


Figure 8.1 *Residual vs. fitted plots for the Galápagos model. The first uses fitted values in the scale of the response while the second uses fitted values in the scale of the linear predictor. The third plot uses response residuals while the first two use deviance residuals.*

There are just a few islands with a large predicted number of species while most predicted response values are small. This makes it difficult to see the relationship between the residuals and the fitted values because most of the points are compressed on the left of the display. Now we try plotting $\hat{\eta}$:

```
plot(residuals(modp) ~ predict(modp,type="link"),
     xlab=expression(hat(eta)),ylab="Deviance residuals")
```

Now the points, shown in the second panel of Figure 8.1, are more evenly spaced in the horizontal direction. We are looking for two main features in such a plot. Is there any nonlinear relationship between the predicted values and the residuals? If so, this would be an indication of a lack of fit that might be rectified by a change in the model. For a linear model, we might consider a transformation of the response, but this is usually impractical for a GLM since it would change the assumed distribution of the response. We might also consider a change to the link function, but often this is undesirable since there are a few choices of link function that lead to easily interpretable models. It is best if a change in the choice of predictors or transformations on these predictors can be made since this involves the least disruption to the GLM. For this particular plot, there is no evidence of nonlinearity.

The variance of the residuals with respect to the fitted values should also be inspected. The assumptions of the GLM would require constant variance in the plot and, in this case, this appears to be the case. A violation of this assumption would prompt a change in the model. We might consider a change in the variance function $V(\mu)$, but this would involve abandoning the Poisson GLM since this specifies a particular form for the variance function. We would need to use a quasi-likelihood GLM described in Section 9.4. Alternatively, we could employ a different GLM for a count response such as the negative binomial. Finally, we might use weights if we could identify some feature of the data that would suggest a suitable choice.

For all GLMs but the Gaussian, we have a nonconstant variance function. However, by using deviance residuals, we have already scaled out the variance function and so, provided the variance function is correct, we do expect to see constant variance in the plot. If we use response residuals, that is $y - \hat{\mu}$, as seen in the third panel of Figure 8.1:

```
plot(residuals(modp,type="response") ~ predict(modp,type="link"),
     xlab=expression(hat(eta)),ylab="Response residuals")
```

We see a pattern of increasing variation consistent with the Poisson.

In some cases, plots of the residuals are not particularly helpful. For a binary response, the residual can only take two possible values for given predicted response. This is the most extreme situation, but similar discreteness can occur for binomial responses with small group sizes and Poisson responses that are small. Plots of residuals in these cases tend to show curved lines of points corresponding to the limited number of observed responses. Such artifacts can obscure the main purpose of the plot. Difficulties arise for binomial data where the covariate classes have very different sizes. Points on plots may represent just a few or a large number of individuals.

Investigating the nature of the relationship between the predictors and the response is another primary objective of diagnostic plots. Even before a model is fit to the data, we might simply plot the response against the predictors. For the Galápagos data, consider a plot of the number of species against the area of the island shown in the first panel of Figure 8.2:

```
plot(Species ~ Area, gala)
```

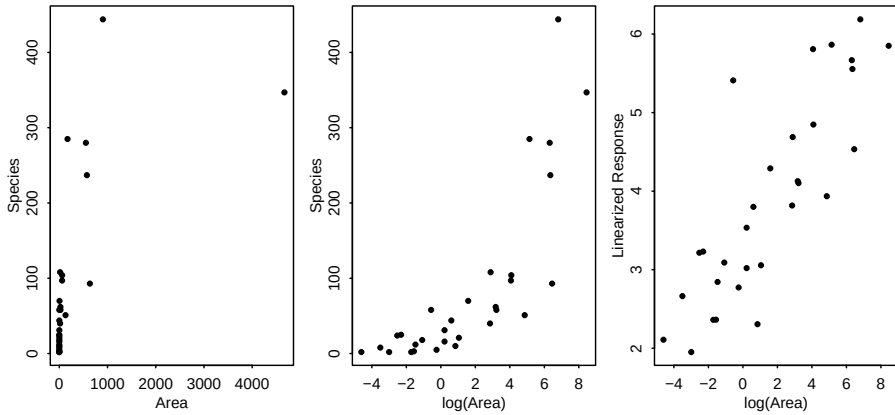


Figure 8.2 *Plots of the number of species against area for the Galápagos data. The first plot clearly shows a need for transformation, the second shows the advantage of using logged area, while the third shows the value of using the linearized response.*

We see that both variables have skewed distributions. We start with a log transformation on the predictor as seen in the second panel of Figure 8.2:

```
plot(Species ~ log(Area), gala)
```

We see a curvilinear relationship between the predictor and the response. However, the default Poisson GLM uses a log link which we need to take into account. To allow for the choice of link function, we can plot the linearized response:

$$z = \eta + (y - \mu) \frac{d\eta}{d\mu}$$

as we see in the third panel of Figure 8.2:

```
mu <- predict(modp,type="response")
z <- predict(modp)+(gala$Species-mu)/mu
plot(z ~ log(Area), gala,ylab="Linearized Response")
```

We now see a linear relationship suggesting that no further transformation of area is necessary. Notice that we used the current model in the computation of z . Some might prefer to use an initial guess here to avoid presuming the choice of model. For this dataset, we find that a log transformation of all the predictors is helpful:

```
modpl <- glm(Species ~ log(Area) + log(Elevation) + log(Nearest) +
  log(Scruz+0.1) + log(Adjacent), family=poisson, gala)
c(deviance(modp), deviance(modpl))
```

```
[1] 716.85 359.12
```

We see that this results in a substantial reduction in the deviance.

The disadvantage of simply examining the raw relationship between the response and the other predictors is that it fails to take into account the effect of the other predictors. Partial residual plots are used for linear models to make allowance for the effect of the other predictors while focusing on the relationship of interest. These can be adapted for use in GLMs by plotting $z - \hat{\eta} + \beta_j x_j$ vs. x_j . The interpretation is the same as in the linear model case. We compute the partial residual plot for the (now logged) area, as shown in the first panel of Figure 8.3:

```
mu <- predict(modpl,type="response")
u <- (gala$Species-mu)/mu + coef(modpl)[2]*log(gala$Area)
plot(u ~ log(Area), gala,ylab="Partial Residual")
abline(0,coef(modpl)[2])
```

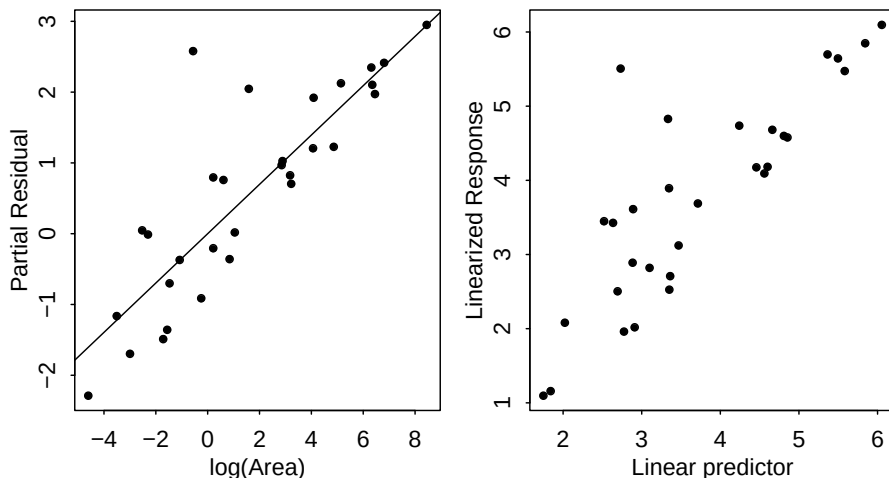


Figure 8.3 A partial residual plot for $\log(\text{Area})$ is shown on the left while a diagnostic for the link function is shown on the right.

In this plot, we see no reason for concern. There is no nonlinearity indicating a need to transform nor are there any obvious outliers or influential points. Partial residuals can also be obtained from `residuals(., type="partial")` although an offset will be necessary if you want the regression line displayed correctly on the plot.

One can search for good transformations of the predictors in nongraphical ways. Polynomial terms or spline functions of the predictors can be experimented with, but generalized additive models, described in Chapter 15, offer a more direct way to discover some good transformations.

The link function is a fundamental assumption of the GLM. Quite often the choice of link function is set by the characteristics of the response, such as positivity, or by ease of interpretation, as with logit link for binomial GLMs. It is often difficult to contemplate alternatives. Nevertheless, it is worth checking to see whether the link assumption is not grossly wrong. Before doing this, it is important to eliminate other simpler violations of the assumptions that are more easily rectified such as outliers or transformations of the predictors. After these concerns have been eliminated, one can check the link assumption by making a plot of the linearized response z against linear predictor $\hat{\eta}$. An example of this is shown in the second panel of Figure 8.3:

```
z <- predict(modpl)+(gala$Species-mu)/mu
plot(z ~ predict(modpl), xlab="Linear predictor",
     ylab="Linearized Response")
```

In this case, we see no indication of a problem.

An alternative approach to checking the link function is to propose a family of link functions of which the current choice is a member. A range of links can then be

fit and compared to the current choice. The approach is analogous to the Box–Cox method used for linear models. Alternative choices are easier to explore within the quasi-likelihood framework described in Section 9.4.

Unusual Points: We have already described the raw material of residuals, leverage and influence measures that can be used to check for points that do not fit the model or influence the fit unduly. Let’s now see how to use graphical methods to examine these quantities.

The QQ plot of the residuals is the standard way to check the normality assumption on the errors typically made for a linear model. For a GLM, we do not expect the residuals to be normally distributed, but we are still interested in detecting outliers. For this purpose, it is better to use a half-normal plot that compares the sorted absolute residuals and the quantiles of the half-normal distribution:

$$\Phi^{-1}\left(\frac{n+i}{2n+1}\right) \quad i = 1, \dots, n$$

The residuals are not expected to be normally distributed, so we are not looking for an approximate straight line. We only seek outliers which may be identified as points off the trend. A half-normal plot is better for this purpose because in a sense the resolution of the plot is doubled by having all the points in one tail.

Since we are more specifically interested in outliers, we should plot the jackknife residuals. An example for the Galápagos model is shown in the first panel of Figure 8.4:

```
halfnorm(rstudent(modpl))
```

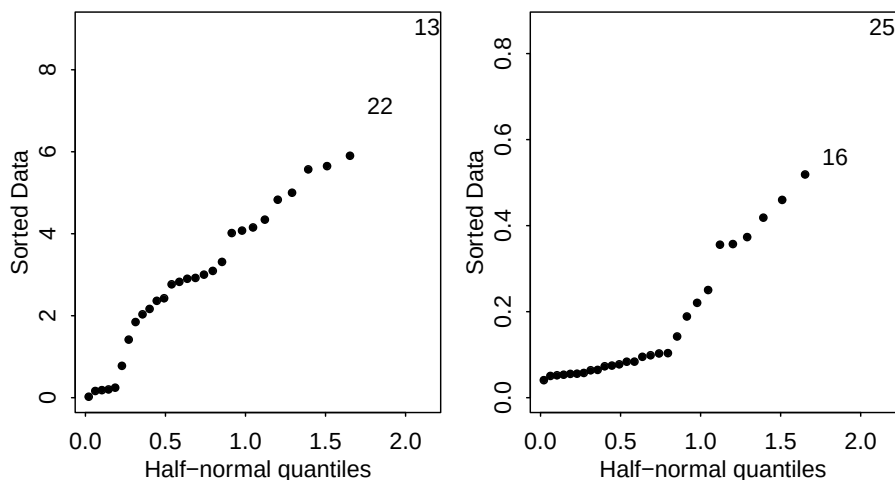


Figure 8.4 *Half-normal plots of the jackknife residuals on the left and the leverages on the right.*

We see no sign of outliers in the plot. The half-normal plot is also useful for positive-valued diagnostics such as the leverages and the Cook statistics. A look at the leverages is shown in the second panel of Figure 8.4:

```
gali <- influence(modpl)
halfnorm(gali$hat)
```

There is some indication that case 25, Santa Cruz island, may have some leverage. The predictor `Scruz` is the distance from Santa Cruz island which is zero for this case. This posed a problem for making the log transformation and explains why we added 0.1 to this variable. However, there is some indication that this inelegant fix may be causing some difficulty.

Moving on to influence, a half-normal plot of the Cook statistics is shown in the first panel of Figure 8.5:

```
halfnorm(cooks.distance(modpl))
```

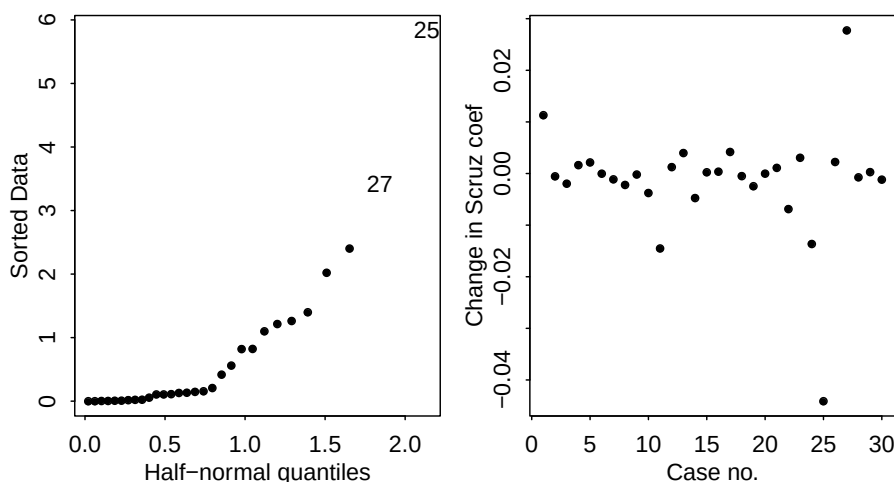


Figure 8.5 Half-normal plot of the Cook statistics is shown on the left and an index plot of the change in the `Scruz` coefficient is shown on the right.

Again we have some indication that Santa Cruz island is influential. We can examine the change in the fitted coefficients. For example, consider the change in the `Scruz` coefficient as shown in the second panel of Figure 8.5:

```
plot(gali$coef[,5],ylab="Change in Scrutz coef",xlab="Case no.")
```

We see a substantial change for case 25. If we compare the full fit to a model without this case, we find:

```
modplr <- glm(Species ~ log(Area) + log(Elevation) + log(Nearest)
+ log(Scruz+0.1) + log(Adjacent), family=poisson, gala, subset=-25)
cbind(coef(modpl),coef(modplr))
```

	[,1]	[,2]
(Intercept)	3.287941	3.050699
log(Area)	0.348445	0.334530
log(Elevation)	0.036421	0.059603
log(Nearest)	-0.040644	-0.052548
log(Scruz + 0.1)	-0.030045	0.015919
log(Adjacent)	-0.089014	-0.088516

We see a sign change for the `Scruz` coefficient. This is interesting since in the full

model, the coefficient is more than twice the standard error way from zero indicating some significance. A simple solution is to add a larger amount, say 0.5, to `Scruz`.

Other than this user-introduced anomaly, we find no difficulty. Using our earlier discovery of the log transformation, some variable selection and allowing for remaining overdispersion, our final model is:

```
modpla <- glm(Species ~ log(Area)+log(Adjacent), family=poisson, gala)
dp <- sum(residuals(modpla,type="pearson")^2)/modpla$df.res
summary(modpla,dispersion=dp)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.2767	0.1794	18.26	< 2e-16
log(Area)	0.3750	0.0326	11.50	< 2e-16
log(Adjacent)	-0.0957	0.0249	-3.85	0.00012

overdispersion parameter = 16.527

n = 30 p = 3

Deviance = 395.543 Null Deviance = 3510.729 (Difference = 3115.186)

Notice that the deviance is much lower and the elevation variable is not used when compared with our model choice in Section 5.1.

This example concerned a Poisson GLM. Diagnostics for binomial GLMs are similar, but see Pregibon (1981) and Collett (2003) for more details.

8.5 Sandwich Estimation

Inference from GLMs depends on the accuracy of the assumptions of the model. Diagnostics are one way in which we can detect violations for these assumptions. Sometimes we can modify the model so that these violations no longer occur or are sufficiently minor to ignore. But sometimes we may not be able or wish to change the model. Instead we would like to modify the inferential procedures so that they are insensitive to these violations. Two common forms of violation are heteroscedascity (unequal variance) and autocorrelation in the response. In GLMs, excepting the Gaussian case, the variance of the response varies as a function of the mean response. By heteroscedascity, we mean the problem where this variation goes beyond what is expected from the mean.

In the standard linear model, ordinary least squares estimation is not optimal in the presence of heteroscedascity and/or correlation in the errors. Nevertheless, it still retains some desirable properties (unbiasedness) and we might retain these estimates especially if we were unsure of a modification that would surely improve them. For GLMs, we also wish to retain the standard MLEs of β because these will still be reasonable. For example, we have seen that these estimators are unchanged in the presence of overdispersion. However, the estimate of $\text{var } \hat{\beta}$ is much more sensitive to these violations of the assumptions. *Sandwich* estimation is a method for obtaining better estimates of $\text{var } \hat{\beta}$ in these circumstances.

We describe how these are applied for standard linear models (LM) to avoid complicating the exposition but the extension to GLMs is straightforward. We have

$$y = X\beta + \varepsilon \quad \text{where} \quad \text{var } \varepsilon = \Omega$$

Using the standard ordinary least squares estimate $\hat{\beta}$ gives

$$\text{var } \hat{\beta} = (X^T X)^{-1} X^T \Omega X (X^T X)^{-1}$$

The usual assumption of independent and identical errors has $\Omega = \sigma^2 I$ which means we reduce to $\text{var } \hat{\beta} = \sigma^2 (X^T X)^{-1}$. But suppose we believe that this assumption has been violated. Let's focus on the heteroscedastic case with no correlation. We might estimate $\Omega = \text{diag}(\omega_1, \dots, \omega_n)$ in order to obtain a better estimate of the variance. Various estimates have been proposed but one that has been shown to behave well in the small sample case is:

$$\omega_i = \frac{\hat{\epsilon}_i^2}{(1 - h_i)^2}$$

for leverages h_i . This allows the computation of a sandwich estimate of the variance of $\hat{\beta}$. The estimator gets its name from the bread, $(X^T X)^{-1}$ and the meat, $X^T \Omega X$.

The estimator can be extended to handle autocorrelation in the response which might be a concern for time-ordered data. We have described the LM solution but the extension to GLMs is straightforward. For more details, see Zeileis (2004).

Sandwich estimators are implemented in the `sandwich` package. We illustrate their application to the Galápagos data using the final model of the previous section:

```
data(gala, package="faraway")
library(sandwich)
modpla <- glm(Species ~ log(Area)+log(Adjacent), family=poisson, gala)
(sebeta <- sqrt(diag(vcovHC(modpla))))
      (Intercept)      log(Area)      log(Adjacent)
      0.181162      0.034877      0.026938
```

Notice that these standard errors for the regression coefficients are quite similar to the previous model where we used a dispersion parameter to account for the known violation of the mean equal to variance assumption of the strictly Poisson GLM. Hence the sandwich estimator also takes care of the overdispersion problem and it is not necessary to introduce a dispersion parameter into the sandwich-based solution. Wald tests of the parameters can be obtained by simply dividing the parameter estimates by their standard errors as before with much the same outcome.

8.6 Robust Estimation

When we compute the MLE for a GLM we are required to solve a set of equations as seen in Section 8.2:

$$\sum_i \gamma_i \frac{(y_i - \mu_i)}{V(\mu_i)} \frac{\partial \mu_i}{\partial \beta_j} = 0 \quad \forall j$$

We have added weights γ_i to give us some additional flexibility. These weights are distinct from previously defined weights w_i and are used for the specific purpose of downweighting extreme observations where y_i is far from μ_i . One problem with the standard MLE is that as a particular y_i becomes more extreme, the influence on $\hat{\beta}$ is unbounded. We want to define γ_i in such a way that each observation has *bounded influence*. Now consider

$$E \gamma \frac{(y - \mu)}{V(\mu)} \frac{\partial \mu}{\partial \beta_j}$$

When $\gamma = 1$ we know from standard likelihood theory that this expectation is equal to zero. We will need γ to be defined as a function of y , x and β to ensure we achieve bounded influence but we still want the expectation to be zero as in the standard case. In other words, we want these to be *conditionally unbiased* given x .

These considerations lead to the definition of the CUBIF (conditionally unbiased influence function) method of robust estimation. Our description has been intuitive but for the full details see Kunsch et al. (1989). This method is implemented in the `robust` package of Wang et al. (2014) which we apply to the same Galápagos model of the previous sections:

```
data(gala, package="faraway")
library(robust)
rmodpla <- glmrob(Species ~ log(Area)+log(Adjacent), family=poisson,
  ↪ data=gala)
summary(rmodpla)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.98321	0.05198	57.4	<2e-16
log(Area)	0.42115	0.00925	45.5	<2e-16
log(Adjacent)	-0.11214	0.00660	-17.0	<2e-16

Robustness weights w.r * w.x:

9 weights are ~ 1. The remaining 21 ones are summarized as

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0879	0.2540	0.3820	0.4340	0.6300	0.8330

This implementation uses a Huber-style influence function. There are various fitting parameters which can be tweaked but we have shown the default choices here.

There are two strategies for using robust models. One approach is to use them as the primary method of estimation. We can use them to make inferences about relationships between the variables and to predict future observations. The drawback is that the statistical methodology to achieve this is more complicated and less developed than for standard GLMs. Progress can be made but it is more difficult. In some applications, a very large number of models will be fitted or future data will be used so that it will be impractical for a human analyst to judge the results. In such circumstances, we may prefer the robust modeling approach as it will be less sensitive to unexpected or aberrant observations.

The other approach is to view the robust fit as an ancillary to the standard GLM fit. We can compare the results of these two fits. In this particular example, we find that the coefficients are not substantially different. We might proceed to make conclusions using the standard fit with the added confidence that these results were not unduly sensitive to some extreme observations. If we had seen a substantially different fit, we would investigate to find the cause of the difference.

It is worth checking which observations received the lowest weighting in the robust model:

```
wts <- rmodpla$w.r
names(wts) <- row.names(gala)
head(sort(wts))
```

Gardner1	Rabida	Bartolome	Seymour	Daphne.Minor
0.087863	0.114472	0.214854	0.216445	0.228807

Our previous diagnostic analyses also detected unusual points but these methods struggle with the problem that such points often influence the fit itself. At most, they

can check the effect of excluding one case whereas robust methods are able to handle multiple unusual cases. In this example, we might examine these cases more closely to see what makes them unusual.

Further Reading: The canonical book on GLMs is McCullagh and Nelder (1989). Other books include Dobson and Barnett (2008), Lindsey (1997), Myers et al. (2002), Gill (2001) and Fahrmeir and Tutz (2001). For a Bayesian perspective, see Dey et al. (2000).

Exercises

- Data is generated from the exponential distribution with density $f(y) = \lambda \exp(-\lambda y)$ where $\lambda, y > 0$.
 - Identify the specific form of $\theta, \phi, a(), b()$ and $c()$ for the exponential distribution.
 - What is the canonical link and variance function for a GLM with a response following the exponential distribution?
 - Identify a practical difficulty that may arise when using the canonical link in this instance.
 - When comparing nested models in this case, should an F or χ^2 test be used? Explain.
 - Express the deviance in this case in terms of the responses y_i and the fitted values $\hat{\mu}_i$.
- The Conway–Maxwell–Poisson distribution has probability function:

$$P(Y = y) = \frac{\lambda^y}{(y!)^v} \frac{1}{Z(\lambda, v)} \quad y = 0, 1, 2, \dots$$

where

$$Z(\lambda, v) = \sum_{i=0}^{\infty} \frac{\lambda^i}{(i!)^v}$$

Place this in exponential family form, identifying all the relevant components necessary for use in a GLM.

- Consider the following distributions and determine whether they can be put in exponential family form. If it is not possible, consider whether there is a special case which follows the form.

- The Uniform distribution:

$$f(y|\alpha, \beta) = \frac{1}{\beta - \alpha} I(y \in [\alpha, \beta])$$

- The Weibull distribution: ($y > 0$)

$$f(y|\alpha, \beta) = \alpha \beta^{-\alpha} y^{\alpha-1} e^{-(y/\beta)^\alpha}$$

- The Gumbel distribution: ($y > 0$)

$$f(y|\alpha, \beta) = \frac{1}{\beta} \exp((y - \alpha)/\beta - \exp((y - \alpha)/\beta))$$

- Consider the Galápagos data and model analyzed in this chapter. The purpose of this question is to reproduce the details of the GLM fitting of this data.
 - Fit a Poisson model to the species response with the five geographic variables as predictors. Do not use the endemics variable. Report the values of the coefficients and the deviance.
 - For a Poisson GLM, derive η , $d\eta/d\mu$, $V(\mu)$ and the weights to be used in an iteratively fit GLM. What is the form of the adjusted dependent variable here?
 - Using the observed response as initial values, compute the first stage of the iteration, stopping after the first linear model fit. Compare the coefficients of this linear model to those found in the GLM fit. How close are they?
 - Continue the iteration to get the next η and μ . Use this to compute the current value of the deviance. How close is this to the deviance from the GLM?
 - Compute one more iteration of the GLM fit, reporting the next calculation of the coefficients and deviance. How close are these to target now?
 - Repeat these iterations a few more times, computing the deviance in each time. Stop when the deviance does not change much. Compare your final estimated coefficients to that produced by the GLM fit.
 - Use your final iterated linear model fit to produce standard errors for the coefficients. How close are these to that produced by the direct GLM fit?
- Again using the Galápagos data, fit a Poisson model to the species response with the five geographic variables as predictors. Do not use the endemics variable. The purpose of this question is to compare six different ways of testing the significance of the elevation predictor, i.e., $H_0 : \beta_{Elev} = 0$. In each case, report the p -value.
 - Use the z -statistic from the model summary.
 - Fit a model without elevation and use the difference in deviances to make the test.
 - Use the Pearson Chi-squared statistic in place of the deviance in the previous test.
 - Fit the Poisson model with a free dispersion parameter as described in Section 5.2. Make the test using the model summary.
 - Use the sandwich estimation method for the standard errors in the original model. Use these to compute z -statistics.
 - Use the robust GLM estimation method and report the test result from the summary.
 - Compare all six results. Pick the best one and justify your choice.

6. The `worldcup` data were collected at the 2010 World Cup. We are interested in modeling the number of shots taken by each player. As goalkeepers do not normally shoot, you should remove them from the dataset. Due to substitution and varying success in the tournament, the number of minutes played by each player is quite variable. For this reason, compute new variables that represent the number of tackles and passes made per 90-minute game.
- (a) Fit a Poisson model with the number of shots as the response and team, position, tackles and passes per game as predictor. Note that time played is a rate variable and should be accounted for as described in Section 5.3. Interpret the effect of tackles and passes on shots.
 - (b) Calculate the leverages for the current model. Report which player has the highest leverage and suggest why this might be so. Make an appropriate plot of the leverages and comment on whether any leverage is exceptional.
 - (c) Compute the change in the regression coefficients when each case is dropped. In particular, examine the change in the tackle coefficient identifying the player which causes the greatest absolute change in this value. What is unusual about this player? Plot the change in the tackle coefficient and determine if any of the values is particularly large.
 - (d) Calculate the Cook Statistics. Which player has the largest such statistic and what is unusual about him?
 - (e) Find the jackknife residuals. Find the player with the largest absolute residual of this kind. How did he come to be the largest?
 - (f) Plot the residuals against the appropriate fitted values. Explain the source of the lines of points appearing on the plot. What does this plot indicate?
 - (g) Make the following three plots:
 - i. Plot raw shots against tackles.
 - ii. Plot shots per game against tackles per game.
 - iii. Plot the linearized response against tackles per game.Make an interpretation of each plot and choose the best one for discovering the relationship between this predictor and the response. Justify your choice.
 - (h) Construct the partial residual plot for tackles and interpret. Is the point on the far right really influential?
 - (i) Make a diagnostic plot to check the choice of the (default) link function.

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